MINI PROJECT – SCHOOL DROPOUT PREVENTION

Cohort – 13

Sl.No	Member Name	Assigned Role
1	Neha Malkud	Documentation
2	Tanvi Jadhav	Documentation
3	DEEKSHITH B K	ML Development
4	Megha Bhalke	ML Development
5	Soniya	Data cleaning
6	Shilpa Shree	Data cleaning
7	Ateeth	FeatureEngineering and data visualization.
8	Rayachoti Rajesh	FeatureEngineering and data visualization.
9	Anjali KJ	Model evaluation and hyperparameter tuning
10	Vijay kumar R K	Data collection
11	Priyadharshan	ML Development
12	Sanchan Chandrashekhar	Data collection
13	Priyanka M Masabinal	ML Development
14	SUDEEPA P	Model evaluation and hyperparameter tuning
15	Vikas S R	Model evaluation and hyperparameter tuning

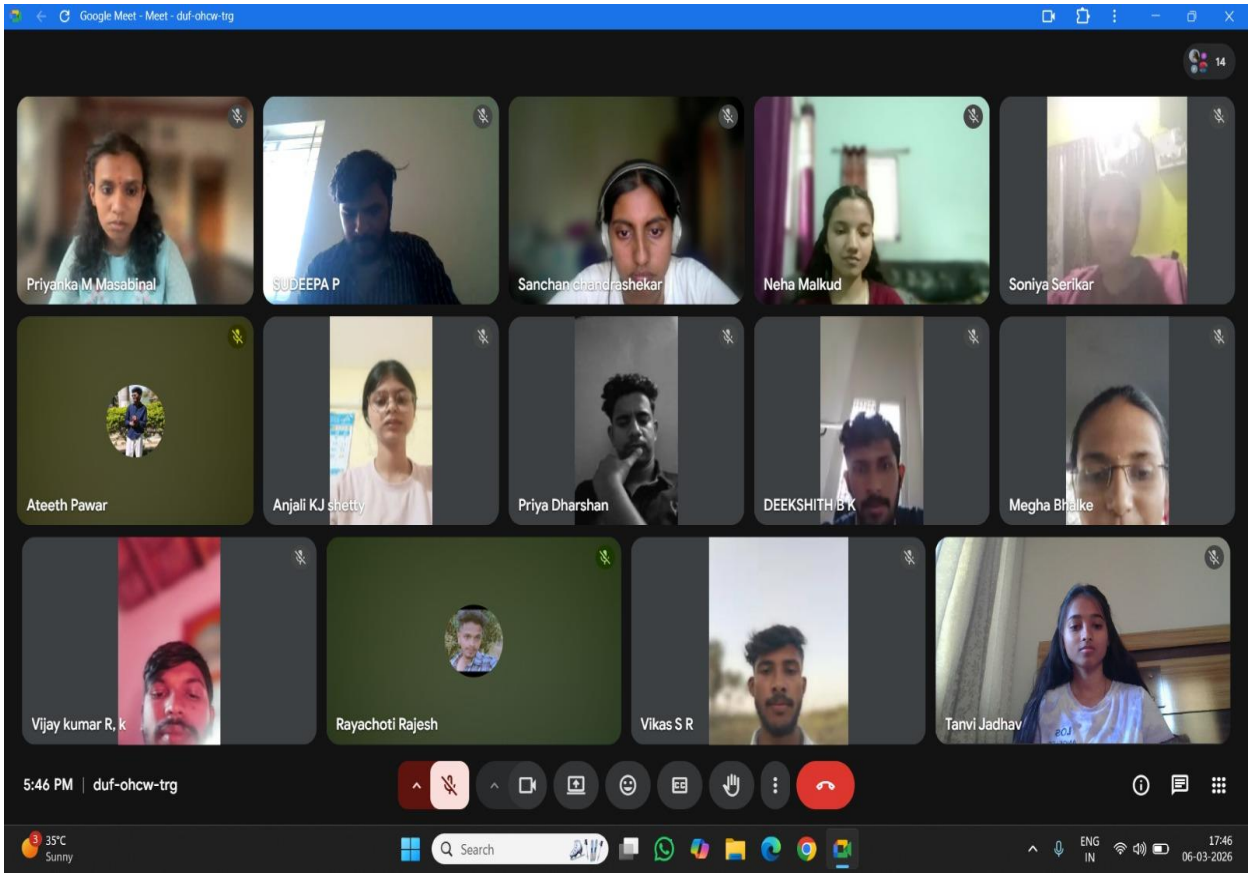
DAY 1 – Role Assignment and Initial Data Structuring

Work done:

- Roles and responsibilities were finalized.
- Dataset structure decided
- Documentation team compiled raw data into Excel.
- Cleaning team reviewed initial dataset for errors.
- Project workflow plan created.
- Source : kaggle

Project initiation meeting

- Platform: Google Meet
- Duration : 1hr
- Screenshot: Attached below



DAY 02 - Dataset Collection and understanding

Work done :

1.Data Collection Team (Vijay Kumar R K, Sanchan Chandrashekhar)

Collected additional datasets related to student demographics and academic performance and Shared dataset files with the entire team.

2. Data Cleaning Team (Soniya, Shilpa Shree)

Shilpa shree did not update on her work contribution .
Checked dataset for missing values and inconsistencies and Identified duplicate entries and incorrect data formats.

3. Feature Engineering and Data Visualization Team (Ateeth, Rayachoti Rajesh)-

not active.

4. ML Development Team (Deekshith B K, Megha Bhalke, Priyadarshan, Priyanka M Masabinal) Priyadarshan was not active.Discussed possible machine learning algorithms suitable for the project.

5. .Model Evaluation & Hyperparameter Tuning – Anjali KJ

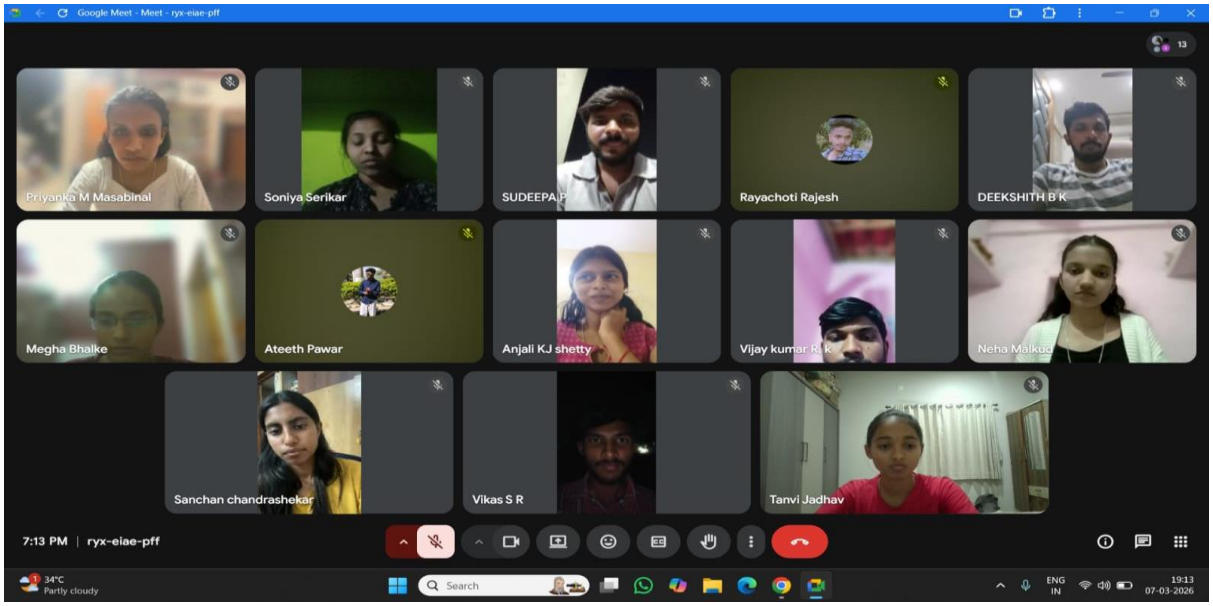
(Sudeepa P and Vikas)- did not interact
Discussed evaluation metrics required for model performance analysis.

6.Documentation Team (Neha Malkud, Tanvi Jadhav)

Updated project documentation with dataset details.

Project initiation meeting

- Platform: Google Meet
- Duration : 1hr
- Screenshot: Attached below



DAY 03 – Dataset finalization and initiation of data cleaning and pre-processing

Work done :

1.Data Collection Team (Vijay Kumar R K, Sanchan Chandrashekhar)
Collected and finalized dataset related to school dropout prevention and shared it with the team.

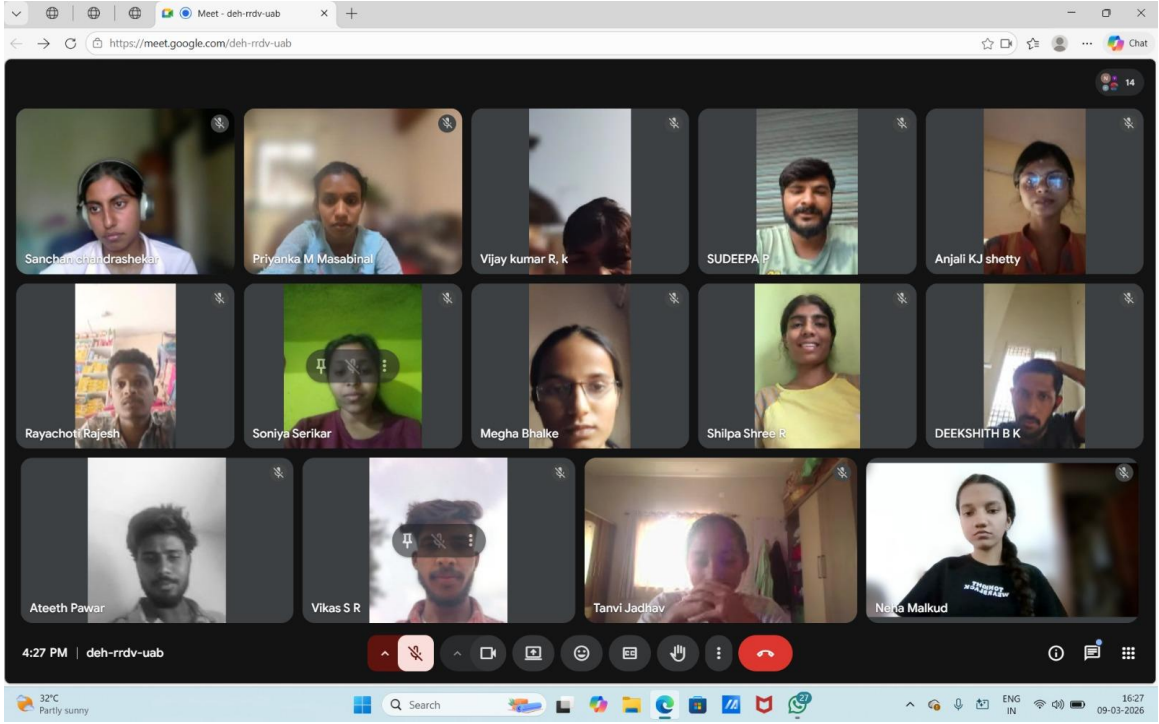
2. Data Cleaning Team (Soniya, Shilpa Shree)
Shilpashree – due to some reasons was not active .
Started data cleaning by identifying missing values, removing duplicates, and organizing the dataset.

3. Feature Engineering and Data Visualization Team (Ateeth, Rayachoti Rajesh) - Did not interact or mentioned about their work regarding mini project.

4. ML Development Team (Deekshith B K, Megha Bhalke, Priyadarshan, Priyanka M Masabinal) Priyadarshan was not active.Discussed possible machine learning algorithms suitable for the project.

5. Model Evaluation & Hyperparameter Tuning – Anjali KJ, Sudeepa P and Vikas Anjali - Worked on iris dataset and went through the theory of support vector

6.Documentation Team (Neha Malkud, Tanvi Jadhav)
Documented project progress, updated daily log, and recorded dataset finalization and team activities.



DAY 04 – Data cleaning and pre processing

Work done :

1.Data Collection Team (Vijay Kumar R K, Sanchan Chandrashekhar)

Assisted the team in their work.

2. Data Cleaning Team (Soniya, Shilpa Shree)

Soniya and Shilpa Shree focused on preparing the dataset for analysis. They examined the raw dataset to identify missing values, duplicate records, and inconsistent entries. After identifying these issues, they removed duplicate records and handled missing values appropriately. As part of preprocessing, they applied feature scaling using a scaler to normalize numerical values so that all features are within a similar range, which helps improve the performance of machine learning algorithms.

3. Feature Engineering and Data Visualization Team (Ateeth, Rayachoti Rajesh)

Ateeth Pawar and Rajesh worked on identifying the most important features that may affect student dropout prediction. They explored the cleaned dataset and analyzed relationships between different attributes such as attendance, academic scores, and family background.

4. ML Development Team (Deekshith B K, Megha Bhalke, Priyadarshan, Priyanka M Masabinal) Priyadarshan was not active.

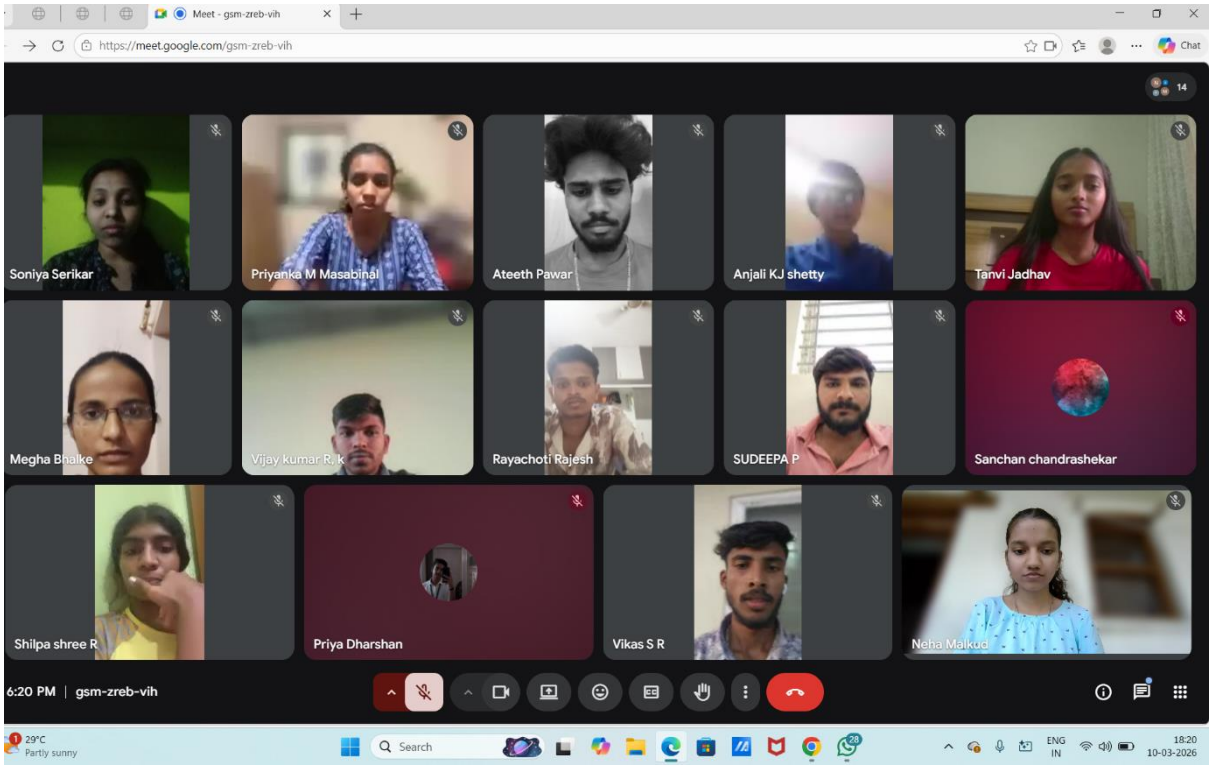
Priyanka,Deekshith,and Megha Bhalke focused on understanding which machine learning algorithms could be suitable for predicting student dropout risk they are individually implementing different models.

5. Model Evaluation & Hyperparameter Tuning – Anjali KJ, Sudeepa P and Vikas

Anjali KJ, Sudeepa P, and Vikas studied various evaluation metrics that will be used to measure the performance of the machine learning models. They explored concepts such as accuracy, precision, recall, and F1-score.

6.Documentation Team (Neha Malkud, Tanvi Jadhav)

Neha and Tanvi were responsible for documenting the progress of the project. They recorded the work completed by each team, maintained the daily project log, and ensured that all steps such as data collection, data cleaning, preprocessing, and team contributions were properly documented for the final project report.



DAY 05- Feature engineering ,visualization and ML development

Work done :

1.Data Collection Team (Vijay Kumar R K, Sanchan Chandrashekhar)

Vijay Kumar R K and Sanchan Chandrashekhar assisted the team by organizing and verifying the dataset used in the project. They ensured that the dataset used for feature engineering and model development contained relevant attributes related to student performance, attendance, and socio-economic background.

2. Data Cleaning Team (Soniya, Shilpa Shree)

Soniya and Shilpa Shree reviewed the cleaned dataset once again to ensure there were no remaining inconsistencies. They verified that missing values were handled correctly and confirmed that the dataset was properly formatted for feature engineering and model training. Their work ensured that the dataset was reliable and ready for further analysis.

3. Feature Engineering and Data Visualization Team (Ateeth, Rayachoti Rajesh)

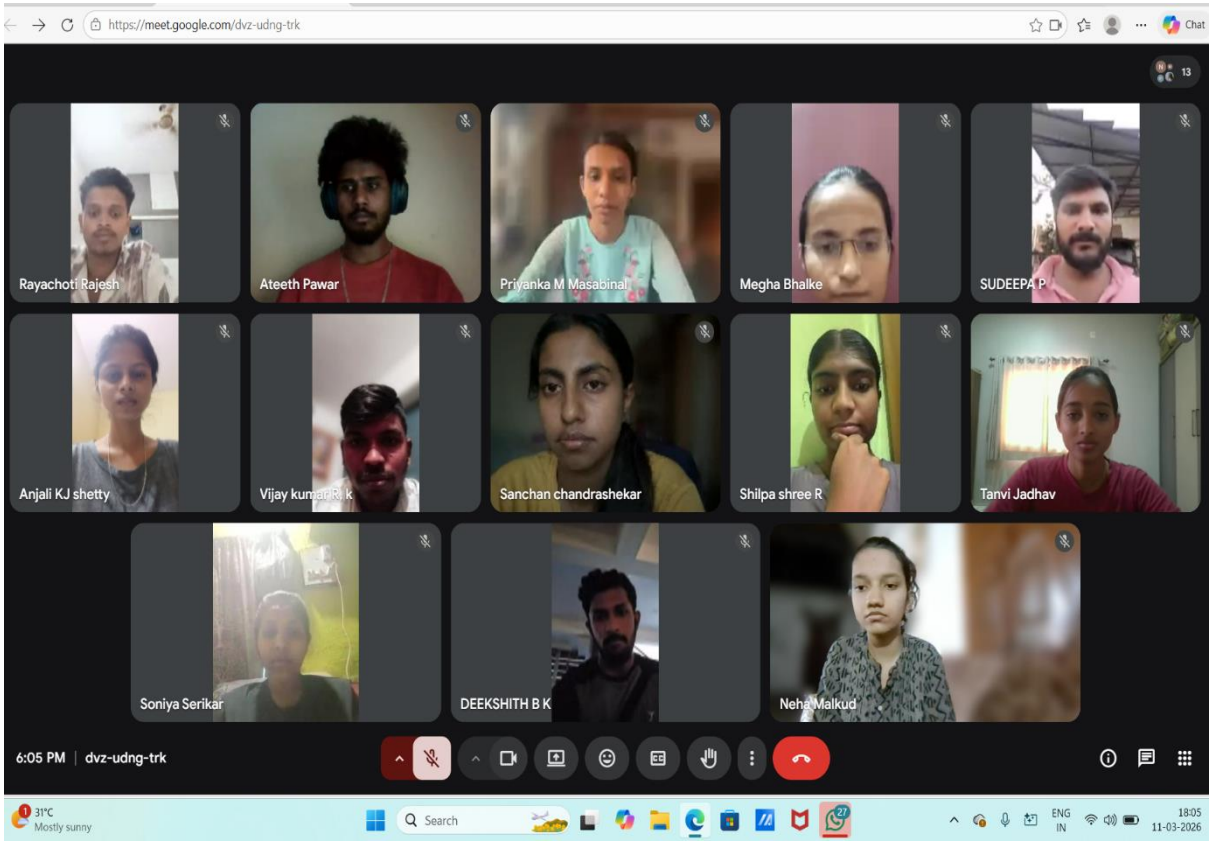
Ateeth and Rayachoti Rajesh focused on creating new features that could improve the prediction of student dropout. They analyzed the dataset and generated derived attributes such as attendance rate and average academic performance. They also created visualizations like bar charts and correlation plots to understand relationships between variables such as attendance, academic scores, and family background.

4. ML Development Team (Deekshith B K, Megha Bhalke, Priyadarshan, Priyanka M Masabinal)

The ML development team worked on building machine learning models to predict student dropout. They implemented classification algorithms such as K-Nearest Neighbors (KNN), Logistic Regression, Decision Tree, and Random Forest. The models were trained using the preprocessed dataset and tested to analyze their prediction performance. The team compared the results of different models to identify the most suitable algorithm for the project.

5. Model Evaluation & Hyperparameter Tuning – Anjali KJ, Sudeepa P and Vikas

Anjali KJ, Sudeepa P, and Vikas S R worked on evaluating the performance of the machine learning models. They calculated evaluation metrics such as accuracy, precision, recall, and F1-score for each model. They also experimented with hyperparameter tuning to improve the performance of the algorithms and identify the most effective model for predicting student dropout risk.

6.Documentation Team (Neha Malkud, Tanvi Jadhav)

Recorded the activities performed by each team during feature engineering, visualization, and model development. They also documented the dataset transformations, machine learning models used, and the evaluation results obtained from different algorithms.